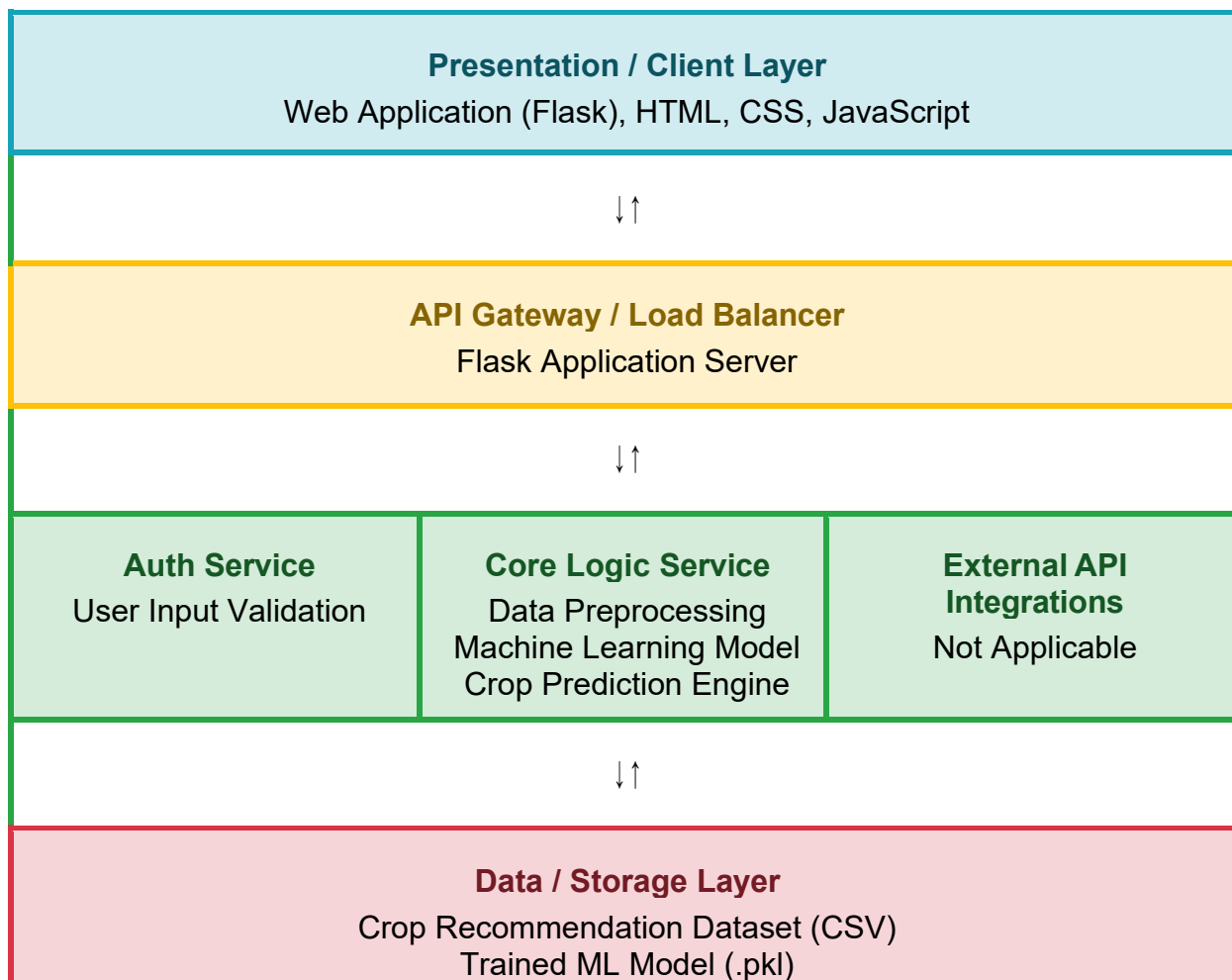


Solution Architecture

Date	27 June 2026
Team ID	
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides the user interface for entering soil and environmental parameters and displaying crop recommendations.	HTML, CSS, JavaScript, Flask
API Gateway	Handles user requests and routes them to the prediction service.	Flask
Core Logic Service	Processes user input, loads the trained ML model, and predicts the most suitable crop.	Python, Scikit-learn, Pandas
Data Layer	Stores the crop dataset and trained machine learning model used for prediction.	CSV Dataset, Joblib (.pkl)